

# DISTINGUISHING HOMEOMORPHIC 4-MANIFOLDS BY SLICING AND THE EXOTIC $S^2 \times S^2$ PROBLEM

BERND J. WUEBBEN

**ABSTRACT.** Lidman and Piccirillo constructed the first pair of closed 4-manifolds with isomorphic integer cohomology rings distinguished by unconstrained knot slicing: spin rational homology 4-spheres  $B$  and  $W$  with  $H_1 = \mathbb{Z}/2$  in which the figure-eight knot is slice and not slice, respectively. We show that the natural upgrade of their theorem — a *homeomorphic* such pair — is equivalent to the exotic  $S^2 \times S^2$  problem: for every admissible Luttinger-surgery variant  $V'$  of their key piece (we classify the admissible parameters),  $W' = V'/\sigma$  is homeomorphic to  $B$  if and only if the symplectic double  $Z' = V' \cup_\sigma V'$  is simply connected, in which case  $Z'$  is an exotic  $S^2 \times S^2$ ; if moreover the piece  $V'$  itself is simply connected, their regluing additionally yields a simply connected exotic  $\mathbb{CP}^2 \# \overline{\mathbb{CP}^2}$ . The proof combines the Hambleton–Kreck classification with a rigidity analysis of the surgery parameters. We then localize what stands between the construction and the goal: building a fully explicit model of the relevant surface-bundle group from the trefoil monodromy, we verify by machine that all 72 uncorrected candidate presentations of  $\pi_1(V')$  collapse to the trivial group, and that this collapse is over-determined — carried entirely by four monodromy relations broken by the surgery tori — so that the value of  $\pi_1(V')$  rests on the based-loop correction words alone. The same tools verify that the group theory of Akhmedov–Park’s claimed exotic  $S^2 \times S^2$  (arXiv:1005.3346) is correct given the presentation of their Lemma 8, isolating the open content of that sixteen-year-old preprint in the geometric derivation of its meridian and framing words. The computation itself — deriving the actual based-loop data and deciding the groups — is carried out in a companion paper.

## 1. INTRODUCTION

One approach to distinguishing smooth structures, going back to Casson, is to find a knot that is smoothly slice in one 4-manifold and not in another with the same underlying topology. Lidman and Piccirillo [LP25] recently ran the first successful version of this program at the level of cohomology rings:

**Theorem 1.1** ([LP25, Theorem 1]). *There are spin rational homology four-spheres  $B$  and  $W$  with  $H_1 = \mathbb{Z}/2$  and isomorphic integer cohomology rings such that the figure-eight knot  $4_1$  is slice in  $B$  but not in  $W$ .*

Here  $B$  is the Kawauchi manifold [Kaw09] and  $W = V/\sigma$ , where  $V$  is a symplectic homology  $S^2 \times D^2$  built from a genus-2 surface bundle  $R$  over a once-punctured torus by two Luttinger surgeries, and  $\sigma$  is a free involution of  $\partial V = S_0^3(Q)$ ,  $Q$  the square knot. The natural strengthening — replace “isomorphic cohomology rings” by “homeomorphic”, producing the first homeomorphic-but-not-diffeomorphic pair distinguished by unconstrained slicing — requires exactly  $\pi_1(W') = \mathbb{Z}/2$  for a variant  $W'$  of the construction. Our first result says this is not an incremental improvement but a genuine obstruction, and explains, we believe, why [LP25] stops at cohomology rings.

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To state it, call a 4-manifold  $V'$  an *admissible variant* of  $V$  if it is obtained from  $R$  by Luttinger surgeries as in Proposition 1.3 below (these are exactly the surgeries on the tori  $T_\alpha, T_\beta$  and parallel copies producing a homology  $S^2 \times D^2$ ; the LP manifold is the case  $(m, n) = (0, 0)$  with no extra tori). Set  $W' = V'/\sigma$  and  $Z' = V' \cup_\sigma V'$ .

**Theorem 1.2** (Reduction). *Let  $V'$  be any admissible variant. Then:*

- (a)  $W' \cong_{\text{homeo}} B$  if and only if  $\pi_1(Z') = 1$ .
- (b) If  $\pi_1(Z') = 1$ , then  $Z'$  is homeomorphic to  $S^2 \times S^2$  and not diffeomorphic to it, and the pair  $(B, W')$  is homeomorphic and non-diffeomorphic, distinguished by the sliceness of  $4_1$ .
- (c) If  $\pi_1(V') = 1$  (which implies  $\pi_1(Z') = 1$ ), then in addition the regluing of [LP25, Theorem 2] applied to  $Z'$  yields a simply connected 4-manifold homeomorphic but not diffeomorphic to  $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$ .

An exotic  $S^2 \times S^2$  — or an exotic  $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$  — is a long-standing open problem; the only claim in the literature, by Akhmedov and Park [AP10], has remained an unpublished preprint since 2010 and is cited as such by the current literature [SS23, BSS24], while [LP25] re-proves the strictly weaker “nonstandard cohomology  $S^2 \times S^2$ ” statement (their Theorem 8) and describes the known examples as “presumably not simply connected”. Theorem 1.2 thus cuts both ways: the slicing upgrade cannot be obtained without solving the exotic  $S^2 \times S^2$  problem inside a  $\sigma$ -symmetric double, and conversely any solution of the  $\pi_1$ -killing problem in the LP frame is worth three theorems at once — strictly dominating the Akhmedov–Park frame, whose payoff is the exotic  $S^2 \times S^2$  alone, at zero marginal cost, since surgeries in  $\text{int } V$  descend to the quotient automatically.

The proof of Theorem 1.2(a) rests on the topological classification of 4-manifolds with finite cyclic fundamental group: by Freedman and Hambleton–Kreck [Fre82, HK88, HK93] (see [Ham08, Theorem 5.1] for the statement in this form, and [BSS24] for its  $b_2 = 0$  case), a closed oriented topological 4-manifold with  $\pi_1 = \mathbb{Z}_n$  is determined up to homeomorphism by its intersection form,  $w_2$ -type, and Kirby–Siebenmann invariant; in particular the smooth spin rational homology 4-sphere with  $\pi_1 = \mathbb{Z}/2$  is unique up to homeomorphism. The input on the smooth side is a rigidity statement for the surgery parameters:

**Proposition 1.3** (Rigidity of the surgery parameters). *Any Luttinger surgeries on  $T_\alpha, T_\beta \subset R$  producing a homology  $S^2 \times D^2$  have coefficient  $\pm 1$ , and the surgery directions equal  $\alpha', \beta'$  modulo fiber classes. Conversely, for every  $m, n \in \mathbb{Z}$  the directions  $\beta' e^m, \alpha' c^n$  with coefficients  $\pm 1$  (and, optionally,  $\pm 1$ -surgeries along parallel Lagrangian copies of  $T_\alpha, T_\beta$  with fiber directions) yield  $H_1(V') = 0$  and preserve: the spin condition, symplecticity,  $\chi = 2$ ,  $H_2(V') = \mathbb{Z}\langle F \rangle$ , the descent of  $\sigma$ , and the hypotheses of [SS23, Theorem 1.4] for the double. In particular [LP25, Lemma 7] holds verbatim for  $W'$ , and the slicing obstruction for  $4_1$  in  $W'$  persists.*

The second half of the paper concerns what stands between the construction and  $\pi_1(V') = 1$ , and makes the obstruction quantitative. In Section 2 we build a fully explicit finite presentation of  $\pi_1(R)$  with no genus-2 mapping class computations: since  $Q$  is the square knot, the closed genus-2 fiber splits along the connect-sum circle and the entire bundle structure is generated by the trefoil monodromy

$$h: x \mapsto y^{-1}, \quad y \mapsto yx$$

on  $F_2 = \langle x, y \rangle$ , together with the handle swap. All structural identities (that  $h$  is the composite of the two standard twists; that it fixes the boundary word  $[x, y]$  exactly; that  $h^6$

is the boundary Dehn twist and  $h^3$  the hyperelliptic involution; that Lidman–Piccirillo’s commutator factorization  $abd^{-1}e^{-1} = (ab)\varphi(ab)^{-1}\varphi^{-1}$  is the literal automorphism identity  $\tilde{\psi}\tilde{\varphi}\tilde{\psi}^{-1}\tilde{\varphi}^{-1} = h * h^{-1}$ ) are certified by machine; the scripts are ancillary files.

With the model in hand, Section 5 reports three computations (GAP 4.16, coset enumeration; all runs reproduce from the ancillary scripts in seconds):

- (E1) All 72 *uncorrected* candidate presentations of  $\pi_1(V')$  — over all sign and ordering conventions, the LP directions, the mixed directions  $m, n \in \{-1, 1\}$ , and the parallel-tori scheme — collapse to the trivial group, while the controls ( $\pi_1(R)$  alone:  $H_1 = \mathbb{Z}^2$ ; either surgery alone:  $H_1 = \mathbb{Z}$ ) do not. If these presentations were correct, LP’s own  $Z$  would already be an exotic  $S^2 \times S^2$ ; they are therefore over-determined, and the failure is localized in exactly four monodromy relations, those carried by transport annuli that the surgery tori must intersect.
- (E2) Dropping the four broken relations outright leaves  $H_1 = \mathbb{Z}^2$ : the corrections are indispensable already for homology, so no certificate assembled from the clean relations can bypass them. The value of  $\pi_1(V')$  — and with it, by Theorem 1.2, the existence of an exotic  $S^2 \times S^2$  — is carried entirely by the based-loop correction words, and a naive enumeration proves nothing in either direction.
- (E3) From the presentation printed in [AP10, Lemma 8], coset enumeration confirms  $\pi_1(M_n^p) = \mathbb{Z}/p$  for  $(n, p) \in \{(1, 1), (2, 1), (3, 1), (1, 2), (1, 3)\}$ : the group theory of [AP10, Theorem 9] is correct *given* Lemma 8. What remains open in [AP10] is precisely the geometric derivation of the six meridian/framing words of their Lemma 7 — the same kind of based-loop data as in (E2), and not machine-checkable from a figure.

We read (E2) as the formal content of the field’s sixteen-year hesitation over [AP10]: at this level of explicitness, nothing short of a genuine based diagram decides the group. The flip side is constructive: the model reduces the obstruction to a finite, well-posed computation — derive the correction words for the broken monodromy relations (four over the embedded curves; a minimal based diagram realizes three) and the two based surgery relations from an octagon picture of the fiber — after which the ancillary harness decides  $\pi_1(V')$ , and with it three open problems, in seconds.

Finally, the geometric dual of  $T_\alpha$  requires a small correction to the proof of [LP25, Lemma 5] (Remark 3.1), which is used throughout; and the based-loop computation posed by (E2) — deriving the actual meridian, lasso and Lagrangian-framing words and deciding the groups — is carried out in the companion paper [W26b].

## 2. THE EXPLICIT MODEL OF $\pi_1(R)$

We recall the construction of [LP25, Section 1]. The 0-surgery  $S_0^3(Q)$  on the square knot  $Q = 3_1 \# \overline{3_1}$  fibers over  $S^1$  with closed genus-2 fiber  $F$  and monodromy conjugate to  $abd^{-1}e^{-1}$ , a positive Dehn twist along each of the chain curves  $a, b, c, d, e$  of [LP25, Figure 1] being denoted by the same letter. Let  $\varphi$  be the involution of  $F$  exchanging  $a \leftrightarrow e$ ,  $b \leftrightarrow d$  and preserving  $c$  setwise. Since  $abd^{-1}e^{-1} = (ab)\varphi(ab)^{-1}\varphi^{-1}$  in the mapping class group, the  $F$ -bundle  $R$  over the once-punctured torus with monodromy  $ab$  along  $\beta$  and  $\varphi$  along  $\alpha$  has  $\partial R = S_0^3(Q)$ .

**Convention 2.1.**  $[u, v] = uvu^{-1}v^{-1}$ . Automorphisms act on the left and compose right-to-left:  $fg$  means “ $g$  first”. Mapping classes act on  $\pi_1$  only up to inner automorphism; every presentation below depends on a choice of automorphism lifts, see Remark 2.6.

**2.1. The trefoil monodromy.** Let  $\Sigma_{1,1}$  be a once-punctured torus with basepoint on the boundary,  $\pi_1(\Sigma_{1,1}) = F_2 = \langle x, y \rangle$ , boundary word  $[x, y]$ , and let the core curves realizing  $x, y$  be  $a \simeq x$  and  $b \simeq y$ , so  $a \cdot b = 1$ . For a suitable orientation convention the twists act by

$$T_a: (x, y) \mapsto (x, yx), \quad T_b: (x, y) \mapsto (xy^{-1}, y).$$

Define  $h := T_a \circ T_b$  ( $T_b$  first).

**Lemma 2.2.**  $h(x) = y^{-1}$ ,  $h(y) = yx$ , and:

- (1)  $h([x, y]) = [x, y]$  exactly (not merely up to conjugacy);
- (2)  $h^6 = \text{conj}_{[x, y]^{-1}}$ , the full boundary Dehn twist;
- (3)  $h^3$  acts on  $H_1$  as  $-\text{id}$ ;
- (4) on  $H_1(\Sigma_{1,1}) = \mathbb{Z}^2$ ,  $h$  acts by  $\begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}$ : determinant 1, trace 1, order 6.

Consequently  $h$  is the fibered monodromy of a trefoil: it is a product of single positive twists along dual non-separating curves, fixes the boundary pointwise (reflected in (1)), and is freely periodic of period 6 with  $h^6$  the boundary twist (2), with the hyperelliptic involution at  $h^3$  (3).

*Proof.* Finite computations in  $F_2$ , certified by the ancillary scripts `monodromy_check2.g` and `model_check3.g`, and easily reproduced by hand. For (1):

$$h([x, y]) = [y^{-1}, yx] = y^{-1} \cdot yx \cdot y \cdot x^{-1}y^{-1} = xyx^{-1}y^{-1} = [x, y].$$

□

**Remark 2.3** (Chirality). Whether  $h$  or  $h^{-1}$  ( $h^{-1}: x \mapsto xy, y \mapsto x^{-1}$ ) is the right-handed trefoil is a convention we will not need:  $Q$  is the connected sum of a trefoil and its mirror, and exchanging the chirality assignment amounts to the handle relabeling  $(x, y) \leftrightarrow (r, s)$  below, under which every construction and every computation in Section 5 is symmetric.

**2.2. The closed fiber and the bundle group.** The fiber of  $Q = 3_1 \# \overline{3_1}$  is the boundary connected sum of the two trefoil fibers, and the closed fiber  $F$  of  $S_0^3(Q)$  splits along the connect-sum circle into two once-punctured tori. Writing  $\pi_1(F) = \langle x, y, r, s \mid [x, y][r, s] \rangle$  with  $(x, y)$  carried by the left handle and  $(r, s)$  by the right, the dictionary with [LP25, Figure 1] is

$$a \sim x, \quad b \sim y, \quad e \sim r, \quad d \sim s, \quad c \sim xr, \quad z \sim ys,$$

where the last two identify the  $\varphi$ -invariant homology classes  $[c] = [a] + [e]$  and  $[z] = [b] + [d]$ ; the specific words  $xr$  and  $ys$  are representative choices (see Remark 2.6 and Section 5). Define

$$\tilde{\varphi}: x \leftrightarrow r, y \leftrightarrow s \quad \text{and} \quad \tilde{\psi} := h * \text{id}: (x, y, r, s) \mapsto (y^{-1}, yx, r, s),$$

lifting  $\varphi$  and  $ab$  respectively;  $\tilde{\psi}$  is well defined on the closed-fiber group because  $h$  fixes the boundary word exactly (Lemma 2.2(1)).

**Lemma 2.4.** (1)  $\tilde{\psi}$  fixes the relator  $[x, y][r, s]$  exactly;  $\tilde{\varphi}$  sends it to its  $[x, y]$ -conjugate. In particular both define automorphisms of  $\pi_1(F)$ .

- (2)  $\tilde{\varphi}^2 = \text{id}$ , and  $\tilde{\psi} \tilde{\varphi} \tilde{\psi}^{-1} \tilde{\varphi}^{-1} = h * h^{-1}: (x, y, r, s) \mapsto (y^{-1}, yx, rs, r^{-1})$ .

*Proof.* Machine-certified (`model_check3.g`); (2) also follows from block algebra:  $\tilde{\varphi}(h * \text{id})^{-1} \tilde{\varphi} = \text{id} * h^{-1}$ , so the commutator is  $(h * \text{id})(\text{id} * h^{-1}) = h * h^{-1}$ . □

Lemma 2.4(2) realizes the mapping-class factorization  $abd^{-1}e^{-1} = (ab)\varphi(ab)^{-1}\varphi^{-1}$  of [LP25] as a literal identity of automorphisms:  $h * h^{-1}$  is the 0-surgery monodromy of the square knot in this model (the left handle twisted by the trefoil, the right by its mirror).

**Proposition 2.5** (The presentation).

$$\pi_1(R) = \langle x, y, r, s, \alpha, \beta \mid [x, y][r, s], \\ \alpha g \alpha^{-1} = \tilde{\varphi}(g), \beta g \beta^{-1} = \tilde{\psi}(g) \ (g \in \{x, y, r, s\}) \rangle.$$

*Proof.*  $R$  fibers over the once-punctured torus  $T_0$  with aspherical fiber  $F$ ; the long exact sequence of the fibration reduces to  $1 \rightarrow \pi_1(F) \rightarrow \pi_1(R) \rightarrow \pi_1(T_0) \rightarrow 1$ , and  $\pi_1(T_0) = F_2\langle\alpha, \beta\rangle$  is free, so the sequence splits and  $\pi_1(R) = \pi_1(F) \rtimes F_2$  with respect to the chosen automorphism lifts  $\tilde{\varphi}, \tilde{\psi}$  of the two monodromies. The displayed presentation is the standard presentation of such a semidirect product.  $\square$

**Remark 2.6** (Lift freedom; where the based-loop indeterminacy lives). A mapping class determines its action on  $\pi_1(F)$  only up to inner automorphisms. Replacing  $\tilde{\varphi}, \tilde{\psi}$  by other lifts changes the presentation of Proposition 2.5 by the substitution  $\alpha \mapsto \alpha w, \beta \mapsto \beta w'$  ( $w, w' \in \pi_1(F)$ ) — an isomorphism of  $\pi_1(R)$ , but one that changes the *words* that geometric objects (meridians, surgery directions) are represented by. All presentations of the surgered manifolds in Section 5 are therefore *candidates*, well defined only once based representatives are fixed; the dependence of  $\pi_1(V')$  on that choice is precisely the point of experiment (E2).

**Remark 2.7** (Convention robustness). Replacing the monodromy lift  $\tilde{\psi}$  by  $\tilde{\psi}^{-1}$  (the holonomy-direction convention) is the substitution  $\beta \mapsto \beta^{-1}$ , which converts the surgery relation  $[z, \alpha]^\varepsilon \beta = 1$  into  $[z, \alpha]^{-\varepsilon} \beta = 1$ : an  $\varepsilon$ -flip, and the computational grid of Section 5 sweeps all  $\varepsilon$ . Together with Remark 2.3, the collapse of the LP cases below is independent of every orientation and composition convention made here; the mixed-direction and parallel-tori cases (whose convention variants differ additionally by word order and conjugation) are run in the conventions stated.

### 3. RIGIDITY OF THE SURGERY PARAMETERS

Recall from [LP25, Section 1] the two Lagrangian tori:  $T_\beta$ , the sub-bundle with fiber  $e$  over  $\beta$ , and  $T_\alpha$ , the sub-bundle with fiber  $c$  over  $\alpha$  (the monodromy  $ab$  fixes  $e$  pointwise;  $\varphi$  preserves  $c$  with orientation). Their meridians are commutators in the complement  $C := R \setminus (T_\alpha \cup T_\beta)$ :  $\mu_{T_\beta} \simeq [z, \alpha'']$  and  $\mu_{T_\alpha} \simeq [d, \beta'']$  (see Remark 3.1), where  $\alpha'', \beta''$  project to  $\alpha, \beta$ . More generally, for the parallel-copy families below, every relevant meridian is such a commutator: the geometric-dual construction applies verbatim to each parallel copy, whose dual sub-bundle torus meets it once and exhibits the meridian as the corresponding commutator.

**Remark 3.1** (The geometric dual of  $T_\alpha$ ). The proof of [LP25, Lemma 5] prints the dual as  $(b, \beta'')$ ; this cannot close up into a torus, since  $ab$  does not preserve  $b$  even homologically (the only twist in  $ab$  affecting  $b$  is  $T_\alpha$ , and  $a \cdot b = 1$ , so  $[ab(b)]$  has  $[a]$ -coefficient  $\pm 1$  in every convention). The curve dual to  $c$  that  $ab$  *does* fix — pointwise, being disjoint from  $a \cup b$ , exactly as [LP25, p. 3] observe for  $e$  — is  $d$ ; the dual is therefore  $T_\alpha^* = (d, \beta'')$ , with  $\mu_{T_\alpha}$  freely homotopic to  $[d, \beta'']$ . Since  $\varphi$  exchanges  $b \leftrightarrow d$ , the slip is a natural one, and no result of [LP25] is affected: their arguments use only that each meridian is a commutator of a fiber curve with a curve projecting to the base. At the  $\pi_1$  level, however, the correction matters, and it is used throughout this paper and the companion [W26b].

**Lemma 3.2.** *Let  $C = R \setminus \nu(\mathcal{T})$  for any finite disjoint union  $\mathcal{T}$  of tori from the two admissible sub-bundle families. Then the inclusion induces an isomorphism*

$$H_1(C) \xrightarrow{\cong} H_1(R) = \mathbb{Z}^2 \langle \alpha', \beta' \rangle.$$

*In particular every fiber class vanishes in  $H_1(C)$ .*

*Proof.* First,  $H_1(R) \cong \mathbb{Z}^2$ : by Proposition 2.5,  $H_1(R)$  is the direct sum of  $\mathbb{Z}^2 \langle \alpha, \beta \rangle$  and the coinvariants of  $H_1(F) = \mathbb{Z}^4$  under the subgroup generated by  $\tilde{\varphi}_*$  and  $\tilde{\psi}_*$ ; the stacked matrix  $(\Phi - I; \Psi - I)$  has Smith normal form  $\text{diag}(1, 1, 1, 1)$  (ancillary script `model_check3.g`), so the coinvariants vanish. (Concretely:  $\Psi - I$  is invertible on the  $(x, y)$ -block, killing  $x, y$ ; then the swap relations kill  $r, s$ .)

For the complement, excision and the Künneth formula for  $(\nu(T), \partial\nu(T)) \cong T^2 \times (D^2, \partial D^2)$  give  $H_1(R, C) = 0$  and  $H_2(R, C) \cong \bigoplus_{T \in \mathcal{T}} \mathbb{Z}$ , generated by the normal disks, whose boundaries are the meridians. The homology sequence of the pair,

$$H_2(R, C) \xrightarrow{\partial} H_1(C) \rightarrow H_1(R) \rightarrow H_1(R, C) = 0,$$

therefore presents  $H_1(R)$  as the quotient of  $H_1(C)$  by the images of the meridians; each meridian is a commutator in  $\pi_1(C)$ , hence trivial in  $H_1(C)$ , so the map  $H_1(C) \rightarrow H_1(R)$  is an isomorphism.  $\square$

*Proof of Proposition 1.3. Forcing.* A Luttinger  $(1/k\text{-log})$  surgery on  $T_\beta$  along an embedded direction, i.e. a primitive class  $p\beta' + qe$  on  $T_\beta$  ( $\gcd(p, q) = 1$ ), fills the boundary of  $\nu(T_\beta)$  so that the class  $\mu_{T_\beta} + k(p\beta' + qe)$  dies (orientation conventions are absorbed into the sign of  $k$ ). By Lemma 3.2, in  $H_1$  this relation reads  $kp\beta' = 0$ , since the meridian is a commutator and  $e$  is a fiber class. Together with the corresponding relation  $k'p'\alpha' = 0$  from  $T_\alpha$ ,

$$H_1(V') = \mathbb{Z}^2 / \langle kp\beta', k'p'\alpha' \rangle = \mathbb{Z}/|kp| \oplus \mathbb{Z}/|k'p'|,$$

which vanishes iff  $|kp| = |k'p'| = 1$ : the coefficients are  $\pm 1$  and the directions have base component  $\pm 1$ , i.e. equal  $\beta', \alpha'$  modulo fiber classes. Purely-fiber directions ( $p = 0$ ) impose no relation and leave  $H_1 = \mathbb{Z}^2$ .

*Realization.* Conversely, for coefficients  $\pm 1$  and directions  $\beta'e^m, \alpha'c^n$  the same computation gives  $H_1(V') = 0$ ; adding  $\pm 1$ -surgeries along parallel copies with fiber directions adds only trivial  $H_1$ -relations and keeps  $H_1(V') = 0$ . The admissible parallel families are constrained by the intersection pattern of the fiber curves: with  $[c] = [a] + [e]$  and  $[z] = [b] + [d]$  one has  $c \cdot e = 0$ ,  $c \cdot d = z \cdot e = d \cdot e = 1$ ,  $z \cdot d = 0$ , so the maximal disjoint families are  $\{c\text{-fibers over } \alpha\text{-parallels}\} \cup \{e\text{-fibers over } \beta\text{-parallels}\}$  (the LP family) or the complementary  $\{z/\alpha\} \cup \{d/\beta\}$  family, but no mixture.

*Preserved structure.* Each direction above is a primitive class realized by an embedded curve on the corresponding Lagrangian torus, so the surgeries are Luttinger surgeries and the result carries a symplectic structure [Lut95, ADK03];  $\chi$  is unchanged. The argument of [LP25, Lemma 5] now applies verbatim:  $H_1(V') = 0$  and  $\partial V'$  connected give  $H_3(V') = 0$  and  $H_2(V')$  free;  $\chi(V') = 2$  gives  $H_2(V') \cong \mathbb{Z}$ , generated by a fiber  $F$  disjoint from all surgery tori;  $F^2 = 0$ ; since  $H_1(V') = 0$  we get  $H_2(V'; \mathbb{Z}/2) \cong \mathbb{Z}/2 \langle F \rangle$ , and evaluation against it detects  $H^2(V'; \mathbb{Z}/2)$ , so  $\langle w_2, F \rangle = F \cdot F = 0$  forces  $w_2(V') = 0$ :  $V'$  is spin. All surgeries take place in  $\text{int } V$ , so  $\partial V' = \partial V = S_0^3(Q)$  and the free involution  $\sigma$  descends; the proof of [LP25, Lemma 7] applies verbatim and  $W' = V'/\sigma$  is a spin rational homology 4-sphere with  $H_1(W') = \mathbb{Z}/2$ .



*Persistence of the obstruction input.* The double  $Z' = V' \cup_{\sigma} V'$  is obtained from the genus-2 bundle  $R \cup_{\sigma} R$  over a genus-2 surface by Luttinger surgeries on disjoint Lagrangian tori; the fiber  $F$  and the section  $\hat{\Gamma}$  coming from a fixed point of  $\varphi$  ([LP25, Lemma 6]) form a hyperbolic pair, and all surgery tori miss a fiber (they lie over curves in the base, which miss a point) and miss the two fixed-point sections (the fiber curves  $c, e, z, d$  and their parallel copies can be isotoped off the two fixed points of  $\varphi$ , exactly as in [LP25, Lemma 6]). These are the hypotheses of [SS23, Theorem 1.4] as used in [LP25, proof of Theorem 1, p. 6], which constrains neither the surgery coefficients nor the directions. Hence the slicing obstruction for  $4_1$  in  $W'$  persists; the details are completed in the proof of Theorem 1.2(b) below.  $\square$

#### 4. THE REDUCTION THEOREM

*Proof of Theorem 1.2. (a).* The double cover  $Z' \rightarrow W'$  gives a short exact sequence  $1 \rightarrow \pi_1(Z') \rightarrow \pi_1(W') \rightarrow \mathbb{Z}/2 \rightarrow 1$ , with  $\pi_1(Z') \hookrightarrow \pi_1(W')$  *injective* by covering theory.

( $\Leftarrow$ ) If  $\pi_1(Z') = 1$  then  $\pi_1(W') \cong \mathbb{Z}/2$ . Now  $W'$  and  $B$  are closed, oriented, *smooth* spin 4-manifolds with  $\pi_1 = \mathbb{Z}/2$  (Proposition 1.3 for  $W'$ ; [Kaw09, LP25] for  $B$ ) and both are rational homology spheres, so both intersection forms on  $H_2/\text{tors}$  are trivial, both  $w_2$ -types are type (II), and both Kirby–Siebenmann invariants vanish (smoothness; alternatively  $\text{KS} \equiv \sigma/8 \pmod{2}$  for spin topological 4-manifolds). By the classification of closed oriented topological 4-manifolds with finite cyclic fundamental group — [HK93, Theorem C]:  $\pi_1$ , the intersection form on  $H_2/\text{Tors}$ , the  $w_2$ -type and KS are a complete set of invariants (extending [HK88, Theorem B] from odd order; see also [Ham08, Theorem 5.1] and [Tei92]) — we get  $W' \cong_{\text{homeo}} B$ . In fact, by the  $b_2 = 0$  case as stated in [BSS24] there are exactly two such topological manifolds for each cyclic group, one spin and one non-spin; the spin one is the spun lens space, so  $W' \cong_{\text{homeo}} B \cong_{\text{homeo}} L_2$ .

( $\Rightarrow$ ) A homeomorphism gives  $\pi_1(W') \cong \pi_1(B) = \mathbb{Z}/2$ , and the injection above forces  $|\pi_1(Z')| = 1$ .

(b). Assume  $\pi_1(Z') = 1$ .

*Step 1:  $Z'$  is homeomorphic to  $S^2 \times S^2$ .*  $Z'$  is closed, simply connected, spin (it double covers the spin manifold  $W'$ ; alternatively glue the spin structures of the two copies of  $V'$ ), with  $\chi(Z') = 2\chi(V') - \chi(S_0^3(Q)) = 4$ , so  $b_2(Z') = 2$ . The classes of the fiber  $F$  and of the closed section  $\hat{\Gamma} = \Gamma \cup_{\sigma} \Gamma$  satisfy  $F^2 = 0$  and  $F \cdot \hat{\Gamma} = 1$ ; since the form is even (spin), a change of basis  $\hat{\Gamma} \mapsto \hat{\Gamma} - \frac{1}{2}(\hat{\Gamma}^2)F$  exhibits the intersection form as the hyperbolic form  $H$ . By Freedman’s classification [Fre82] (even forms are realized by a unique topological manifold),  $Z' \cong_{\text{homeo}} S^2 \times S^2$ .

*Step 2:  $Z'$  is not diffeomorphic to  $S^2 \times S^2$ .* This is the argument of [LP25, Theorem 8], which is  $\pi_1$ -agnostic:  $R \cup_{\sigma} R$  is a non-trivial genus-2 surface bundle over a genus-2 surface, hence aspherical, hence minimal and neither rational nor ruled, so its Kodaira dimension satisfies  $\kappa \neq -\infty$ ; Luttinger surgery preserves  $\kappa$  [HL12], and  $Z'$  is minimal (it is spin), so  $\kappa(Z') \neq -\infty$ . Since  $\kappa$  is a diffeomorphism invariant [Li06] and  $\kappa(S^2 \times S^2) = -\infty$ ,  $Z'$  is not diffeomorphic to  $S^2 \times S^2$ .

*Step 3: the pair  $(B, W')$ .* By (a),  $W' \cong_{\text{homeo}} B$ . The knot  $4_1$  is slice in  $B$  by construction [Kaw09, LP25]. It is not slice in  $W'$ : following [LP25, Section 2], since  $W'$  is spin the relative Rokhlin theorem [Klu21, Theorem 2] and  $\text{Arf}(4_1) = 1$  rule out a null-homologous slice disk; a homologically essential slice disk makes the 0-trace  $X_0(4_1)$  embed in  $W'$  with  $H_2$ -essential image, producing a square-zero essential torus  $T \subset W'$  (cap a Seifert surface with the disk); since  $X_0(4_1)$  is simply connected,  $T$  lifts to an essential square-zero torus in  $Z'$ ; and Proposition 1.3 verifies for  $Z'$  the hypotheses of [SS23, Theorem 1.4], which forbids

such a torus. Sliceness of  $4_1$  distinguishes the smooth structures, so  $W'$  is homeomorphic and not diffeomorphic to  $B$ .

(c). Assume  $\pi_1(V') = 1$ . The double  $Z'$  and the reglued manifold  $Z'' = V' \cup_{f \circ \sigma} V'$  (with  $f$  the boundary self-homeomorphism of [LP25, Figure 3]) are unions of two copies of  $V'$  along  $S_0^3(Q)$ ; by van Kampen each fundamental group is a quotient of  $\pi_1(V') * \pi_1(V') = 1$ , so  $\pi_1(Z') = \pi_1(Z'') = 1$  and parts (a)–(b) apply. (This is where the hypothesis  $\pi_1(V') = 1$  is genuinely used: the regluing twists the van Kampen data by  $f_*$ , and no implication from  $\pi_1(Z') = 1$  alone is claimed; see Remark 4.1.) The framing-parity change makes  $Z''$  non-spin with  $b_2 = 2$  and signature 0, so its intersection form is  $\langle 1 \rangle \oplus \langle -1 \rangle$  and Freedman gives  $Z'' \cong_{\text{homeo}} \mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$ . The relative invariants  $\Psi_{V', t_{\pm}}$  are non-zero exactly as in [LP25, Lemma 9]:  $Z'$  is symplectic and its canonical class evaluates  $\pm 2$  on a fiber disjoint from the surgery tori — Luttinger surgery does not change this pairing [ADK03, HL12] — so [OS04] applies as there (cf. [Thu76]). By [LP25, Lemma 10] the mixed invariant of  $Z''$  for the line  $\text{span}\{F\}$  is non-zero; since  $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$  (indeed  $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2 \# D$  for any homology sphere  $D$ ) admits, for every choice of line, a square-zero splitting along  $S^2 \times S^1$  forcing all mixed invariants to vanish ([LP25, proof of Theorem 2], using  $HF_{\text{red}}(S^2 \times S^1) = 0$ ),  $Z''$  is not diffeomorphic to  $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$ .  $\square$

**Remark 4.1.** The implication  $\pi_1(V') = 1 \Rightarrow \pi_1(Z') = 1$  is immediate from van Kampen; we do not know whether the converse holds in this setting, since in the amalgam  $\pi_1(Z') = \pi_1(V') *_{\pi_1(S_0^3(Q))} \pi_1(V')$  the edge maps need not be injective. The distinction affects only part (c) — parts (a) and (b) use  $\pi_1(Z')$  alone, while the regluing twists the van Kampen data by  $f_*$  and is controlled here through  $\pi_1(V')$  — and it is immaterial in practice: any certification of simple connectivity in this frame proceeds by computing  $\pi_1(V')$ , which is what the harness of Section 5 does.

**Corollary 4.2.** *The homeomorphic upgrade of [LP25, Theorem 1] within the LP frame is gated: it holds for some admissible variant if and only if some admissible double  $Z'$  is an exotic  $S^2 \times S^2$ . Conversely, exhibiting  $\pi_1(V') = 1$  for any admissible variant yields simultaneously an exotic  $S^2 \times S^2$ , a homeomorphic-but-not-diffeomorphic pair distinguished by unconstrained slicing, and a simply connected exotic  $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$ .*

## 5. COMPUTATIONAL EXPERIMENTS

All computations are coset enumerations and abelianizations in GAP 4.16 [GAP]; the scripts are provided as ancillary files,<sup>1</sup> and every run below reproduces in seconds on a laptop (the enumeration cap is  $4 \times 10^5$  cosets; a run that exceeds it is reported as *blowup* — not visibly trivial, never “nontrivial”).

**5.1. Candidate presentations.** By Proposition 2.5 and Remark 2.6, a presentation of  $\pi_1(V')$  requires, beyond the exact bundle relations, based representatives for the two surgery relations. The *uncorrected candidates* take the naive words suggested by the free homotopy types: filling relations

$$[z, \alpha]^{\varepsilon_1} \cdot \beta r^m = 1, \quad [w, \beta]^{\varepsilon_2} \cdot \alpha (xr)^n = 1,$$

with  $z \in \{ys, sy\}$ ,  $\varepsilon_i \in \{\pm 1\}$ , mixed-direction exponents  $m, n$  as in Proposition 1.3, and, in the parallel-tori scheme, the additional fiber-direction relations  $[z, \alpha]^{\varepsilon_3} r = 1$ ,  $[w, \beta]^{\varepsilon_4} (xr) = 1$ .

<sup>1</sup>pi1\_grid.g, pi1\_v2b.g, ap\_check.g, monodromy\_check2.g, model\_check3.g; also available at <https://github.com/bwuebben/exotic-s2xs2>.



The grid of (E1) takes  $w = y$ , the dual printed in [LP25]; the corrected dual  $w = s$  of Remark 3.1 is run separately.

**5.2. (E1) The uncorrected candidates collapse universally.** Over all 8 conventions for the LP surgeries, the 32 mixed-direction cases ( $m, n \in \{-1, 1\}$ ), and the 32 parallel-tori cases, *every one of the 72 uncorrected candidate groups coset-enumerates to the trivial group* (scripts `pi1_grid.g`, `pi1_v2b.g`; under a second), while the controls do not (the bundle group alone has  $H_1 = \mathbb{Z}^2$ , and either surgery alone leaves  $H_1 = \mathbb{Z}$ ). This collapse cannot be valid: were it, the original LP manifold would already satisfy  $\pi_1(V) = 1$  and, by Theorem 1.2,  $Z$  would be an exotic  $S^2 \times S^2$  — which [LP25] does not claim. The candidates are therefore over-determined, and the failure is localizable a priori. A monodromy relation  $\beta g \beta^{-1} = \tilde{\psi}(g)$  is carried by a transport annulus  $g \times \beta$ ; in the surgered manifold the relation holds only if the annulus avoids the surgery tori. Intersection numbers in  $R$  decide this: an annulus over one base curve meets a sub-bundle torus over the transverse base curve above their intersection point, and meets a torus over the *same* base curve along it; in both cases the count is the intersection number of the fiber curves. Using  $[c] = [a] + [e]$ ,  $[z] = [b] + [d]$ :

| relation carried over $\alpha$ or $\beta$     | obstruction                                                  |
|-----------------------------------------------|--------------------------------------------------------------|
| $x (\sim a), r (\sim e)$ over $\alpha, \beta$ | $a \cdot c = a \cdot e = e \cdot c = 0$ : clean              |
| $y (\sim b)$ over $\beta$                     | $b \cdot c = 1$ : crosses $T_\alpha$                         |
| $y$ over $\alpha$                             | $b \cdot c = 1$ : crosses $T_\alpha$                         |
| $s (\sim d)$ over $\alpha$                    | $d \cdot c = 1$ : crosses $T_\alpha$                         |
| $s$ over $\beta$                              | $d \cdot e = 1, d \cdot c = 1$ : crosses $T_\beta, T_\alpha$ |

Exactly the four relations for  $y \sim b$  and  $s \sim d$  are broken; in  $V'$  they hold only with insertions of conjugates of words carried by the glued-in tori ( $c$ -,  $e$ -,  $\alpha'$ -,  $\beta'$ -words). The four relations for  $x$  and  $r$  are clean. (The count is that of membranes over the *embedded* curves. Like the correction words themselves, it is diagram data: in the companion's diagram, whose based base loops meet the embedded curves minimally, only three relations acquire corrections, each at a single crossing.)

**5.3. (E2) The corrections decide the group.** Dropping the four broken relations outright leaves a group with  $H_1 = \mathbb{Z}^2$  (script `pi1_v2b.g`): the broken relations are indispensable already for  $H_1(V') = 0$ , so no surjection from a clean-relations group onto the actual  $\pi_1(V')$  can certify anything — the clean group is not even a homology-correct approximation. The value of  $\pi_1(V')$  — and with it, by Theorem 1.2, the existence of an exotic  $S^2 \times S^2$  — is therefore carried entirely by the based-loop correction words, with no robustness in either direction: a naive enumeration that exceeds the coset cap (a *blowup*) proves nothing, and nothing short of a genuine based diagram decides the group. That computation is carried out in [W26b].

**5.4. (E3) Verification of the Akhmedov–Park group theory.** The manifold of [AP10] is built from the same  $\mathbb{Z}/2$ -germ: their model surface  $(\Sigma_2 \times \Sigma_3)/\mathbb{Z}_2$  fibers over  $\Sigma_2$  with monodromy the same two-fixed-point involution class as  $\varphi$ . Their Theorem 9 asserts  $\pi_1(M_n^p) = \mathbb{Z}/p$  from the presentation printed in their Lemma 8. Coset enumeration from that presentation (script `ap_check.g`) confirms

$$\pi_1(M_n^p) = \mathbb{Z}/p \quad \text{for} \quad (n, p) \in \{(1, 1), (2, 1), (3, 1), (1, 2), (1, 3)\}.$$

Thus the group theory of [AP10, Theorem 9] is correct *given* their Lemma 8, and the entire content of the sixteen-year-old open status of [AP10] is the geometric derivation of the

six meridian/Lagrangian-framing triples of their Lemma 7 (their fifth and sixth, involving a path that is not a sub-bundle coordinate, being the visibly delicate ones). One cannot machine-check a figure; the referee’s task for [AP10] and the based-loop computation posed by (E2) are the same task.

## 6. DISCUSSION

**6.1. The companion computation.** The model of Section 2 reduces the obstruction to a finite, fully specified computation: derive, from an explicit based picture of  $F$  carrying the curves  $a, \dots, e, z$  and the basepoint, the actual correction words for the broken monodromy relations and the based surgery data, in the style of [AP10, Lemma 7] or Baldridge–Kirk [BK08]. Because the crossing pattern of Section 5 is fixed by the intersection combinatorics, a single based diagram parameterizes the entire admissible family of Proposition 1.3 at once. This computation is carried out in the companion paper [W26b], from a rotation-symmetric octagon model in which every monodromy lift is genuinely based and every meridian, lasso and Lagrangian-framing word is derived mechanically from finite combinatorial certificates; the resulting presentations are proved complete and decided by coset enumeration. The outcome is that  $\pi_1(V') = 1$  for *all nine* cells  $m, n \in \{-1, 0, 1\}$  of the mixed-direction family computed there — including the Lidman–Piccirillo surgery itself — so that, by Theorem 1.2, the double  $Z$  of [LP25] is an exotic  $S^2 \times S^2$ , the pair  $(B, W)$  is homeomorphic and non-diffeomorphic, distinguished by the sliceness of  $4_1$ , and the regluing yields a simply connected exotic  $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$  — and likewise for every admissible variant. Seven cells are decided by coset enumeration. The remaining two, whose enumerations blow past  $10^8$  cosets — and which a 56-CPU-hour search had shown to have no nontrivial finite quotient of order at most  $10^5$ , as, it turns out, befits the trivial group — are decided by Knuth–Bendix completion, whose confluent rewriting systems reduce every generator to the identity: enumeration blowup proves nothing in either direction, exactly as (E2) warned. We refer to [W26b] for the verification apparatus supporting that computation.

**6.2. An ungated rung.** Theorem 1.2 does not gate every strengthening of [LP25]. Their Theorem 3 produces a 4-manifold  $A$  homeomorphic to  $\mathbb{CP}^2 \# 5\overline{\mathbb{CP}}^2$  in which the non-vanishing of the mixed invariant for the line  $\text{span}\{F\}$  survives; a constrained non-slicing statement on  $A$  — no genus-one knot with non-trivial Arf invariant bounds a disk in a class  $d$  with  $d \cdot F \neq 0$  and  $d^2 \geq 0$  — appears provable with the tools already in [LP25, SS23, MMP24], while the unconstrained statement is plausibly false (the classes invisible to a line-based invariant,  $d \cdot F = 0$  or  $d^2 < 0$ , are exactly where Norman disks live in the standard manifold). We leave this to future work.

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BERND J. WUEBBEN, NEW YORK, NY, [wuebben@gmail.com](mailto:wuebben@gmail.com)